

Multi-Stage Deepfake Detection with Dual Explainability: Spatial and Frequency Domain Analysis

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Abstract—The rapid advancement of generative artificial intelligence has made synthetic media increasingly indistinguishable from authentic content, posing significant threats to digital trust, media integrity, and public security. Existing deepfake detectors treat this problem as a binary classification task—real versus fake—without differentiating between the distinct generative technologies responsible for the manipulation, thereby limiting their forensic utility and providing no interpretable evidence of the underlying generative process.

This paper proposes a multi-stage deepfake detection system comprising two main branches integrated under a unified web platform. The image branch employs a two-stage hierarchical ensemble of three deep learning architectures to first distinguish authentic from artificially generated facial images, and then identify whether a detected fake originates from a generative adversarial network or a diffusion model. A dual explainability module accompanies every decision by producing an ensemble-consistent gradient-based spatial activation map highlighting forgery regions alongside a frequency-domain heatmap that reveals supportive spectral indicators of the generative process. The audio branch extracts cepstral and spectral features from speech signals and classifies them as genuine or synthetic using a hybrid convolutional-recurrent architecture, evaluated on a standard anti-spoofing benchmark.

Preliminary dataset analysis on a balanced corpus of 51,000 facial images sampled from a large-scale demographically annotated collection motivates the proposed pipeline. Quantitative detection performance, explainability quality, and platform latency will be reported in full during the implementation phase.

The proposed platform constitutes, to the best of our knowledge, the first system to jointly offer multi-stage hierarchical detection, three-model ensemble voting, dual spatial-frequency explainability, and hybrid audio deepfake detection within a single deployable tool, providing actionable forensic intelligence across both image and audio modalities.

Index Terms—deepfake detection, generative adversarial networks, diffusion models, ensemble learning, explainable AI, Grad-CAM, FFT frequency analysis, audio deepfake, Mel spectrogram, MFCC, multimodal detection

I. PROJECT GOALS AND OBJECTIVES

The primary goal of this project is to design and implement an end-to-end multimodal platform for deepfake analysis across image and audio modalities, with a strong emphasis on transparency, interpretability, and forensic actionability. The specific objectives are:

- 1) **Multi-Stage Image Classification:** Develop a two-stage hierarchical classifier that first distinguishes real facial images from Artificial Intelligence (AI)-generated ones, then identifies whether a fake image was produced by a Generative Adversarial Network (GAN) or a Diffusion Model (DM), providing generative-provenance intelligence.
- 2) **Ensemble Inference:** Train three complementary architectures (EfficientNet-B4, ResNet-50, and Vision Transformer) independently for each classification stage and combine their predictions via majority-vote ensemble to improve robustness against domain shift and reduce individual-model failure modes.
- 3) **Dual Explainability (XAI):** For every input image, produce two complementary visual explanations: (a) an ensemble-consistent Gradient-weighted Class Activation Map (Grad-CAM) heatmap localising the spatial forgery artifacts in the RGB domain, derived by averaging the attribution maps of all models that agree with the majority-vote decision, and (b) a Fast Fourier Transform (FFT) frequency heatmap revealing supportive spectral indicators left by the generative model, enabling interpretation by non-expert end users.
- 4) **Audio Deepfake Detection:** Implement an independent audio branch that extracts Mel-Frequency Cepstral Coefficient (MFCC) and Mel spectrogram features from

speech signals and classifies them as real or synthetic using a hybrid Convolutional Neural Network–Long Short-Term Memory (CNN-LSTM) architecture, achieving competitive performance on the ASVspoof 2019 logical-access benchmark.

- 5) **Unified Web Platform:** Deploy all components under a single interactive interface (Gradio) with separate tabs for image and audio analysis, displaying detection results, confidence scores, and explainability visualisations within a target latency of 5–10 seconds per query.
- 6) **Benchmark Evaluation:** Report accuracy, F1-score, Area Under the Receiver Operating Characteristic Curve (AUC-ROC), and per-class performance for each stage and each individual model, comparing against state-of-the-art hierarchical detectors including the Guarnera et al. baselines [4], [5] and AWARE-NET [10].

II. INTRODUCTION

Deepfake technology—the synthesis of hyper-realistic human faces, voices, and videos using deep generative models—has evolved from a research novelty into a pervasive societal challenge. The proliferation of Generative Adversarial Networks (GANs) [1] and, more recently, Denoising Diffusion Probabilistic Models (DDPMs) [2] has dramatically lowered the barrier to creating synthetic media that deceives both human observers and automated forensic systems. High-profile incidents, including politically manipulated video of world leaders and AI-cloned voice fraud costing millions of dollars [26], underscore the urgency of reliable, interpretable, and deployable detection tools.

a) Motivation: Current deepfake detectors predominantly frame the problem as binary classification: real or fake [3]. This formulation ignores a forensically critical distinction: GAN-generated images often carry periodic grid-like artifacts in the frequency domain arising from transposed convolution upsampling [17], whereas diffusion-generated images frequently exhibit smoother and more radially symmetric spectral attenuation patterns [6], [7]. A detector unaware of the generative provenance cannot explain *why* an image is fake, severely limiting its utility in legal, journalistic, and security contexts. Furthermore, a significant portion of published detectors rely on a single model architecture, rendering them brittle under distribution shift to unseen generators [8]. The absence of interpretable outputs further erodes trust among non-expert end users, who cannot verify or challenge an automated decision.

In the audio domain, the threat is equally urgent. Advances in text-to-speech (TTS) and voice conversion (VC) systems have enabled the creation of synthetic voices that are perceptually indistinguishable from genuine speech [22]. Existing audio deepfake detectors, however, are rarely integrated with image-based systems into a unified deployment, leaving a critical gap for real-world multimedia forensics.

b) Previous Work and Research Gap: Guarnera et al. introduced a three-level hierarchical ResNet-based pipeline

that classifies images as real, GAN-generated, or diffusion-generated with over 97% accuracy [4], [5]. While this work constitutes the closest prior art to our image branch, it employs a single architecture with no ensemble strategy, provides no explainability visualisations, and does not expose frequency-domain evidence. Frequency-aware approaches such as FreqNet [8] demonstrate that enforcing high-frequency learning substantially improves generalisation across unseen generators, yet they neither integrate visual explanations nor support multi-stage classification. Explainability work combining Grad-CAM with deepfake detection [14], [15] addresses spatial artifacts only and does not examine the frequency domain. In the audio modality, Mel-spectrogram convolutional pipelines [21] achieve competitive performance on ASVspoof benchmarks [22], and hybrid CNN-LSTM architectures [24] combining MFCC and spectrogram features further improve robustness, yet no prior publicly available work unifies audio detection with an image-based multi-stage system in a single deployable platform.

To our knowledge, no prior system simultaneously provides: (i) multi-stage hierarchical classification (Real $\rightarrow$ Fake $\rightarrow$ GAN/DM); (ii) three-model majority-vote ensemble; (iii) dual spatial-frequency explainability (ensemble-consistent Grad-CAM + FFT heatmap); (iv) hybrid CNN-LSTM audio deepfake detection with MFCC and Mel spectrogram features; and (v) a unified interface with independent per-modality analysis tabs. This five-way combination constitutes the research gap that the present work addresses.

c) Methodology: The proposed system consists of two main branches integrated under a Gradio-based web platform. The *image branch* implements a two-stage ensemble classifier backed by EfficientNet-B4 [18], ResNet-50 [19], and a Vision Transformer (ViT) [20], augmented with ensemble-consistent Grad-CAM and FFT visualisation modules. The image branch will be trained and evaluated on a balanced 51,000-image corpus sampled from the AI-Face dataset [25], a million-scale demographically annotated collection accepted at CVPR 2025, covering real faces alongside faces generated by ten GAN architectures and eight Diffusion Model variants. The *audio branch* extracts MFCC and Mel spectrogram features from speech signals and feeds them to a hybrid CNN-LSTM classifier, trained and evaluated on the ASVspoof 2019 logical-access partition [22].

d) Results: Detailed quantitative results will be reported in the GP2 implementation phase; Section IV-F provides the expected performance targets and evaluation protocol.

e) Conclusion: The proposed system advances the state of the art by unifying multi-stage hierarchical detection, ensemble robustness, dual-domain explainability, and audio deepfake detection under a single platform, providing actionable forensic intelligence across both image and audio modalities.

f) Paper Structure: The remainder of this paper is organised as follows. Section III reviews related work across all relevant subfields and identifies the research gap. Section IV

describes the proposed methodology in detail. Section IV-F outlines expected results and evaluation metrics.

III. RELATED WORK

A. Review and Analysis of Related Work

Multi-Stage Hierarchical Detection. Guarnera et al. [4] proposed the first hierarchical pipeline classifying images across three levels—real vs. AI-generated, GAN vs. DM, and specific architecture identification—achieving over 97% accuracy using ResNet models on a 14-class dataset. The same group later extended this work [5] with a larger 83,000-image corpus covering nine GAN architectures and four diffusion models, maintaining similar accuracy. Both works confirm the viability of hierarchical classification but provide no ensemble strategy and no explainability outputs, limitations that the present work directly addresses.

Ensemble-Based Detection. Ensemble combinations of complementary architectures consistently outperform single-model detectors. Coccomini et al. [9] combined EfficientNet with ViT for video deepfake detection, demonstrating that architecturally diverse models provide complementary spatial and temporal coverage. AWARE-NET [10] introduced a learnable two-tier weighting scheme for an ensemble of Xception, Res2Net, and EfficientNet-B7, reaching an AUC of 99.22% on FaceForensics++. DFCon [11], a competitive submission to the IEEE SP Cup 2025, achieved strong results by ensembling MaxViT, CoAtNet, and EVA-02 with supervised contrastive learning. Majority-vote inference, adopted in the present work, is preferred over learned weighted averaging at this stage because it requires no validation set for weight calibration, is fully interpretable, and has been shown to perform comparably to more complex ensemble schemes when constituent models achieve similar individual accuracies [9], [11].

Vision Transformer for Deepfake Detection. Wang et al. [13] provide a comprehensive survey of 14 ViT-based deepfake detectors, noting that ViT’s global self-attention captures long-range patch-level inconsistencies that local CNN filters miss. Luo et al. [12] proposed FA-ViT, which freezes pre-trained ViT weights and trains lightweight adaptive modules to expose forgery features, achieving 93.83% AUC on Celeb-DF.

Frequency Domain Analysis. Durall et al. [17] established that GAN upsampling layers often produce periodic spectral artifacts detectable via Fast Fourier Transform (FFT), with grid-like patterns in the frequency domain arising from transposed convolution operations. Tan et al. [8] built FreqNet, a detector applying convolutional layers jointly on FFT phase and amplitude spectra, surpassing prior frequency-based methods by 9.8% in mean accuracy on 17 unseen GANs. Wang et al. [16] demonstrated that fusing spatial and frequency-domain features outperforms either modality alone, providing the theoretical motivation for our dual-domain XAI module.

Detection of Diffusion-Generated Images. Corvi et al. [6] were the first to expose the forensic traces specific to diffusion models, demonstrating that GAN-trained detectors generalise poorly to DM images due to their distinct spectral signatures, which tend to exhibit smoother and more radially

symmetric attenuation rather than the grid-like patterns often found in GAN outputs. Ricker et al. [7] further confirmed that DM-generated images produce fewer detectable artifacts than GANs and that detectors must be retrained on DM data to achieve reliable detection, providing additional theoretical grounding for treating GAN and DM detection as separate classification tasks.

Explainable AI for Deepfake Detection. Selvaraju et al. [14] introduced Grad-CAM, which uses gradients flowing into the final convolutional layer to produce class-discriminative localisation maps without requiring architectural modifications. Venkateswarulu and Srinagesh [15] applied Grad-CAM to deepfake detection within an EfficientNet-LSTM framework, linking activation patterns to forgery regions around eyes, hair edges, and skin boundaries.

Audio Deepfake Detection. Synthetic speech detection has received sustained attention following the introduction of the ASVspoof benchmark series [22], which evaluates countermeasure systems against TTS and VC attacks under the logical-access scenario. Fathan et al. [21] demonstrated that Mel-spectrogram image-based CNN classifiers achieve an Equal Error Rate (EER) of 2.8% on the ASVspoof 2019 evaluation set, establishing Mel spectrogram representations as highly effective features for anti-spoofing. Beyond single-feature approaches, hybrid CNN-LSTM architectures [24] that jointly model Mel-Frequency Cepstral Coefficients (MFCCs) and spectrogram images have been shown to reach 94.7% accuracy and 97.3% AUC-ROC on the Fake-or-Real dataset, outperforming either feature set in isolation. The complementarity of MFCCs and Mel spectrograms is well-understood: MFCCs provide compact cepstral representations of the vocal tract transfer function that capture phoneme-level artifacts introduced by TTS systems, while Mel spectrograms preserve the full time-frequency structure of the signal, revealing synthesis-specific discontinuities, missing micro-prosodic variation, and unnatural harmonic distributions [21], [24]. The CNN component of the hybrid architecture learns local time-frequency patterns from the spectrogram, while the LSTM component models temporal dependencies across frames, capturing the unnatural smoothness in prosodic contours that characterise synthetic speech. The proposed audio branch adopts this CNN-LSTM design, motivated by its demonstrated superiority over purely CNN-based or LSTM-based alternatives.

Multimodal Detection Platforms. The FaceForensics++ dataset [3], containing over 1.8 million manipulated images, remains the canonical benchmark for face forgery detection. Despite the clear benefit of providing both image and audio analysis capabilities, no prior publicly available system offers multi-stage image classification alongside audio deepfake detection within a single platform with independent per-modality analysis.

Dataset: AI-Face. The image experiments in this work use a balanced sample drawn from the AI-Face dataset [25], a million-scale demographically annotated collection accepted at CVPR 2025. AI-Face aggregates faces from real-image sources (including FFHQ, IMDB-WIKI, and real frames from

FaceForensics++, DFDC, DFD, and Celeb-DF-v2) alongside faces generated by ten GAN architectures (including AttGAN, StyleGAN, ProGAN, and StarGAN) and eight Diffusion Models (including Stable Diffusion 1.5, DALL-E 2, and Midjourney), totalling over 2.1 million images across 37 generation methods [25]. A balanced 51,000-image sub-corpus (17,000 real, 17,000 GAN-generated, 17,000 diffusion-generated) was prepared by uniformly sampling equal numbers of GAN-generated and DM-generated images from the respective subsets of the full collection, and is used in accordance with the AI-Face CC BY-NC-ND 4.0 licence.

B. Summary of Related Work

Table I summarises the most relevant prior works and contrasts them with the proposed system across five dimensions: methodology, result, modality, explainability, and audio support.

C. Significance of Work

The significance of the proposed system lies in five contributions that, taken together, have not appeared in any single prior work:

- 1) **Hierarchical multi-class detection** that goes beyond binary real/fake classification to identify the specific generative technology (GAN or DM), providing actionable forensic intelligence directly useful in legal, journalistic, and security contexts.
- 2) **Ensemble robustness** through three architecturally diverse models—CNN-based EfficientNet-B4, residual ResNet-50, and attention-based ViT—whose complementary inductive biases reduce individual model failure modes and improve generalisation under domain shift.
- 3) **Dual-domain explainability** combining an ensemble-consistent spatial Grad-CAM (showing *where* the model detected manipulation in the RGB image) with FFT frequency heatmaps (revealing supportive spectral indicators of how the generative fingerprint may manifest in the spectral domain). The Grad-CAM explanation is derived by averaging the normalised attribution maps of all constituent models whose predictions agree with the majority-vote decision, ensuring that the visual explanation reflects the full ensemble outcome rather than the behaviour of any single model. Frequency-domain patterns are treated as supportive forensic indicators; their discriminative reliability across all generator architectures will be assessed empirically in GP2.
- 4) **Enhanced audio deepfake detection** via a hybrid CNN-LSTM architecture that jointly exploits MFCC and Mel spectrogram features, capturing both phoneme-level cepstral artifacts and temporal prosodic inconsistencies introduced by TTS and VC systems, evaluated on the standard ASVspoof 2019 benchmark.
- 5) **Unified platform for independent per-modality analysis** that deploys image and audio deepfake detection under a single interactive interface with separate analysis tabs, addressing the growing threat of coordinated

multimedia manipulation where both visual and auditory synthetic content appear together.

IV. APPROACH AND METHODOLOGY

A. System Architecture Overview

The proposed system is structured as two parallel and independent detection branches—an image branch and an audio branch—both served through a unified Gradio-based web platform with separate analysis tabs. The overall pipeline is illustrated in Fig. 1.

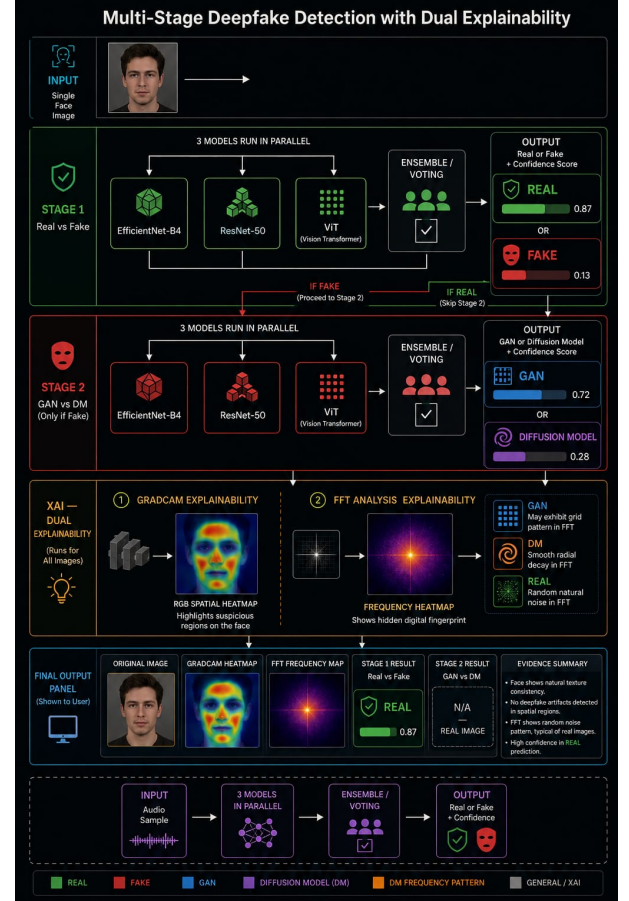


Fig. 1: Overview of the proposed multi-stage deepfake detection system. The image branch (top) performs two-stage ensemble classification with dual XAI outputs. The audio branch (bottom) extracts audio features for CNN-LSTM classification. Both branches operate independently and are served through a unified Gradio platform with separate analysis tabs.

B. Image Branch

1) *Stage 1: Real vs. Fake Classification*: The input facial image is independently forwarded through three pre-trained and fine-tuned models:

- **EfficientNet-B4** [18]: captures fine-grained spatial forgery artifacts via compound-scaled depthwise separable convolutions, providing strong performance on high-resolution facial images with moderate computational cost.

TABLE I: Summary of Related Work and Comparison with Proposed System

Work	Methodology	Result	Modality	XAI	Limitation
Guarnera et al. [4] (2024)	3-level ResNet hierarchy	97%+ acc.	Image only	No	No ensemble; no XAI; no audio
Guarnera et al. [5] (2024)	Extended hierarchy, 83k images	97%+ acc.	Image only	No	No ensemble; no XAI; no audio
Corvi et al. [6] (2023)	Forensic trace analysis for DMs	DM vs. GAN	Image only	No	No classification pipeline
FreqNet [8] (2024)	FFT phase + amplitude CNN	+9.8% mean acc.	Image only	No	No multi-stage; no audio
AWARE-NET [10] (2025)	Learnable 2-tier ensemble	AUC 99.22%	Image only	No	No multi-stage; no audio
FA-ViT [12] (2023)	Adaptive ViT forgery modules	AUC 93.83%	Image only	No	No ensemble; no audio
Fathan et al. [21] (2022)	Mel-spectrogram CNN	EER 2.8%	Audio only	No	No image branch; no XAI
CNN-LSTM [24] (2025)	MFCC + spectrogram CNN-LSTM	94.7% acc.	Audio only	No	No image branch; no multi-stage
Proposed	2-stage ensemble + ensemble Grad-CAM + FFT + CNN-LSTM audio	TBD (GP2)	Image + Audio	Yes	—

- **ResNet-50** [19]: extracts residual features through skip connections, demonstrating robustness to compression and resizing artifacts commonly encountered in real-world deepfake distribution channels.
- **Vision Transformer (ViT)** [20]: models long-range patch-level inconsistencies through global self-attention, capturing structural incoherence across distant facial regions that local CNN receptive fields cannot observe.

Each model produces a binary probability vector $[p_{\text{real}}, p_{\text{fake}}]$. Majority-vote ensemble aggregates the three predictions: the ensemble outputs *Fake* if at least two of the three models independently predict fake; otherwise it outputs *Real*. Majority voting is adopted over learned weighted averaging at this stage because it requires no additional validation data for weight calibration, is interpretable, and has been shown to perform comparably to more complex ensemble schemes when constituent models achieve similar individual accuracies [9], [11]. If the ensemble outputs *Fake*, the image proceeds to Stage 2; otherwise the pipeline terminates and returns *Real* with its confidence score.

2) Stage 2: GAN vs. Diffusion Model Classification:

The same three architectures, independently retrained on GAN/DM-labelled binary data (34,000 images: the GAN-generated and DM-generated subsets of the 51,000-image corpus), classify the fake image to determine its generative origin. Majority voting again determines the final label, GAN or DM, along with per-class confidence bars displayed on the platform interface.

3) *Dual Explainability Module*: For every query image the platform renders three panels side by side:

- 1) **Original Image**: the unmodified input for reference.
- 2) **Ensemble Grad-CAM Heatmap** [14]: each of the three constituent models (EfficientNet-B4, ResNet-50, and ViT) independently generates a Grad-CAM attribu-

tion map from its final convolutional or attention layer. Only the maps produced by models whose individual predictions agree with the final majority-vote decision are retained. These supporting maps are normalised to $[0, 1]$ and averaged to yield a single ensemble-consistent localisation map overlaid on the RGB image:

$$G_{\text{ensemble}} = \frac{1}{|S|} \sum_{m \in S} G_m \quad (1)$$

where G_m denotes the normalised attribution map of supporting model m , and $S \subseteq \{\text{EfficientNet-B4, ResNet-50, ViT}\}$ contains only the models that voted in agreement with the ensemble decision ($|S| \geq 2$ by the majority-vote rule). This design ensures that the visual explanation faithfully reflects the evidence that drove the ensemble outcome rather than the behaviour of any single sub-model. For ViT, Grad-CAM is applied via GradCAM on the last transformer encoder block’s attention-weighted output projection, which has been shown to produce well-localised attribution maps for vision transformers [13]. Based on prior work in frequency-domain forensics [6], [17] and Grad-CAM-based deepfake detection [15], we anticipate that activations for authentic images will be diffuse and low-magnitude; for GAN-generated fakes, they are expected to concentrate around eyes and hair boundaries where transposed convolution artifacts are strongest; and for diffusion-generated fakes, they are expected to highlight over-smoothed skin regions characteristic of denoising score matching. These qualitative patterns will be verified experimentally in GP2.

- 3) **FFT Frequency Heatmap**: a two-dimensional magnitude spectrum is computed by applying a 2-D FFT to

the luminance channel of the image, followed by centre-shifting and log-scale normalisation:

$$M(u, v) = \log(1 + |\mathcal{F}\{I_L\}(u, v)|) \quad (2)$$

where $\mathcal{F}\{I_L\}$ denotes the 2-D discrete Fourier transform of the luminance channel I_L ; `fftshift` is applied before visualisation to place the DC component at the image centre. Real images typically exhibit an approximately isotropic $1/f$ spectral decay. GAN-generated images often exhibit periodic spectral structures associated with transposed convolution upsampling [17], while diffusion-generated images frequently display smoother and more radially symmetric attenuation patterns [6], [7]. These frequency-domain characteristics are treated as supportive forensic indicators rather than definitive proof of a particular generation method; their discriminative reliability across all generator architectures in the AI-Face corpus will be assessed empirically during GP2.

4) *Image Dataset*: Image experiments use a balanced 51,000-image corpus sampled from the AI-Face dataset [25]: 17,000 real faces, 17,000 GAN-generated faces, and 17,000 diffusion-model-generated faces, with pre-defined 80%/20% train/test partitions. AI-Face is a million-scale demographically annotated collection accepted at CVPR 2025, aggregating faces from real-image sources (including FFHQ, IMDB-WIKI, and real frames from FaceForensics++, DFDC, DFD, and Celeb-DF-v2) alongside faces generated by ten GAN architectures and eight Diffusion Model variants across 37 total generation methods [25]. All images are resized to 224×224 pixels and normalised using ImageNet mean and standard deviation statistics prior to model input.

5) *Image Training Protocol*: All image models are initialised with ImageNet pre-trained weights and fine-tuned for 20 epochs using the Adam optimiser with an initial learning rate of 10^{-4} and cosine annealing scheduling. The batch size is set to 32. Data augmentation during training includes random horizontal flipping, colour jitter (brightness ± 0.2 , contrast ± 0.2 , saturation ± 0.2), and Gaussian blur (kernel size 3, $\sigma \in [0.1, 2.0]$). Stage-1 models are trained on all 51,000 images under a real/fake binary label; Stage-2 models are trained independently on the 34,000-image GAN/DM subset.

C. Audio Branch

1) *Motivation and Feature Design*: Synthetic speech produced by modern TTS and voice conversion systems introduces subtle artifacts that are not reliably captured by any single acoustic feature type. Two complementary feature representations are therefore extracted from each audio signal:

- **Mel-Frequency Cepstral Coefficients (MFCCs)**: 40 MFCC coefficients are computed per frame (25 ms window, 10 ms hop, $n_{\text{mels}} = 40$) using a triangular filterbank aligned to the perceptual Mel scale. MFCCs provide a compact cepstral encoding of the vocal-tract transfer function; TTS systems frequently introduce smoothed or over-averaged cepstral trajectories that differ statistically from natural speech coarticulation patterns.
- **Mel Spectrogram**: A full Mel spectrogram is computed with 128 Mel filter banks (25 ms window, 10 ms hop) and converted to decibel scale. Unlike MFCCs, the Mel spectrogram preserves the complete time-frequency structure of the signal, making visible synthesis-specific discontinuities, missing micro-prosodic variation between adjacent phonemes, and unnatural harmonic distributions that characterise current TTS and VC systems.

To ensure consistent feature dimensions across variable-length utterances, all audio signals are normalised to a fixed duration prior to feature extraction. Clips shorter than the target duration are zero-padded at the end, while clips exceeding it are truncated to the predefined length. This standardisation guarantees that MFCC and Mel spectrogram tensors maintain uniform dimensions for CNN-LSTM training and inference. The exact target duration will be selected based on the distribution of utterance lengths in the ASVspoof 2019 training partition and will be reported in GP2.

2) *CNN-LSTM Architecture*: A hybrid Convolutional Neural Network–Long Short-Term Memory (CNN-LSTM) classifier is adopted, following the architecture family demonstrated by [24] to achieve 94.7% accuracy and 97.3% AUC-ROC on the Fake-or-Real dataset. The design rationale exploits a natural division of labour between the two components:

- The **CNN sub-network** treats the Mel spectrogram as a 2-D image and applies successive convolutional and pooling layers to extract local time-frequency patterns associated with synthesis artifacts: spectral gaps, unnatural harmonic regularities, and phase discontinuities.
- The **LSTM sub-network** receives the CNN-extracted frame-level feature vectors and models sequential temporal dependencies across the utterance. This allows the model to detect unnaturally smooth prosodic contours, missing co-articulatory transitions, and the temporal over-regularity that distinguishes TTS output from spontaneous human speech.

The two feature types (MFCCs and Mel spectrogram) are processed in parallel streams that are concatenated before the final fully connected classification head, producing a binary real/synthetic decision with a scalar confidence score. SpecAugment [23]—consisting of random frequency masking and time masking applied to the Mel spectrogram during training—is used as the primary data augmentation strategy to improve robustness to channel mismatch conditions [21].

3) *Audio Dataset and Evaluation Protocol*: Audio experiments use the ASVspoof 2019 logical-access (LA) partition [22], which contains bonafide speech samples alongside spoofing attacks covering TTS and voice conversion systems (attack conditions A01–A19 spanning training, development, and evaluation partitions [22]). The standard ASVspoof evaluation metric is the Equal Error Rate (EER), defined as the operating point at which the false acceptance rate equals the false rejection rate. The target EER on the ASVspoof 2019 LA evaluation set is $\leq 5\%$, a threshold consistent with competitive published systems [21], [24]. Note that this

target is deliberately conservative relative to the 2.8% EER of Fathan et al. [21], as our hybrid CNN-LSTM architecture is trained from random initialisation rather than using pre-trained acoustic models, and exact masking parameters will be tuned on the development set in GP2. Note also that recent state-of-the-art systems on ASVspoof 2019 LA achieve EER below 1%; the 5% target is therefore a minimum bar, and we will report the actual achieved EER alongside published baselines in GP2.

4) *Audio Training Protocol*: The CNN-LSTM model is trained from random weight initialisation for 30 epochs using the Adam optimiser (learning rate 5×10^{-4} , batch size 64) with a cosine annealing schedule. SpecAugment [23] is applied online during training following the LibriSpeech Double (LD) policy of Park et al. [23]: $m_F = 2$ frequency masks each up to $F = 27$ Mel bins wide, and $m_T = 2$ time masks each up to $T = 100$ frames wide, are randomly applied per sample; exact masking parameters will be tuned on the ASVspoof 2019 development set during GP2. Binary cross-entropy is used as the loss function with label smoothing ($\epsilon = 0.1$) to reduce over-fitting to specific attack characteristics.

D. Unified Web Platform

All components are deployed under a single Gradio application with two primary tabs: *Image Analysis* and *Audio Analysis*.

- The **Image Analysis** tab accepts a user-uploaded facial image and returns: (a) Stage-1 real/fake decision with per-model and ensemble confidence bars; (b) if fake, the Stage-2 GAN/DM decision with confidence; and (c) the three-panel ensemble Grad-CAM and FFT heatmap visualisations.
- The **Audio Analysis** tab accepts a user-uploaded audio clip and returns: (a) the real/synthetic classification decision; (b) a confidence score; and (c) the Mel spectrogram visualisation of the input signal, allowing users to visually inspect the time-frequency representation alongside the model's decision.

The target end-to-end inference latency is approximately 5–10 seconds per query on commonly available hardware, including Google Colab T4 GPUs and standard consumer CPUs. The exact latency will depend on model loading, explainability generation, and the execution environment used during deployment, and will be measured and reported in GP2.

E. Location and Safety Considerations

All experiments are conducted on Google Colab using T4 GPU within a controlled research environment. The image corpus is drawn from the AI-Face dataset [25], which was constructed under institutional ethical oversight at Purdue University using publicly released real-image sources, and is used in accordance with its CC BY-NC-ND 4.0 licence exclusively for non-commercial academic research. No personally identifiable information beyond what is contained in the publicly released dataset is collected or stored during platform demonstrations. The ASVspoof 2019 audio dataset is

similarly sourced from consented speakers under the VCTK corpus license. The proposed detection platform is designed exclusively as a forensic and verification tool; the authors acknowledge that deepfake detection systems carry dual-use risk and are committed to responsible disclosure of results and limitations in accordance with institutional research ethics guidelines and Jordan University of Science and Technology research integrity policies.

F. Expected Results and Outputs

As an initial feasibility validation, a ResNet-50 baseline was trained on a 10,000-image binary subset (4,000 real, 4,000 fake) and evaluated on a held-out 2,000-image test set (1,000 real, 1,000 fake). The model achieved 99.75% peak accuracy and 99.60% final accuracy after 10 epochs, with only 8 misclassified samples out of 2,000. Training dynamics and classification performance are shown in Figs. 2 and 3. These results confirm that the AI-Face corpus is well-suited for deep learning classification and establish a strong single-model upper bound that motivates the following GP2 targets.

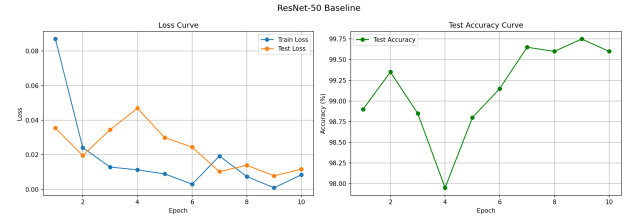


Fig. 2: ResNet-50 baseline training and test loss/accuracy curves over 10 epochs, demonstrating stable convergence on the binary real/fake classification task.

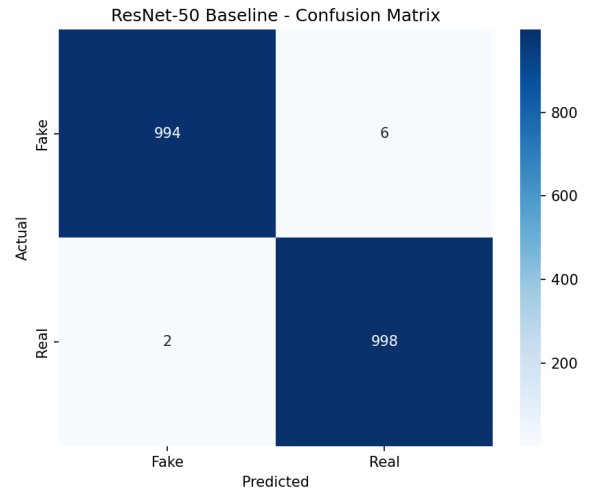


Fig. 3: Confusion matrix of the ResNet-50 baseline on the 2,000-image test set, achieving 99.60% accuracy with only 8 misclassified samples out of 2,000 (6 false negatives, 2 false positives).

Based on this baseline and the ensemble improvements demonstrated by Coccomini et al. [9] and DFCon [11], we

anticipate the following outcomes for the GP2 implementation phase:

- **Stage 1 (Real vs. Fake):** Ensemble accuracy $\geq 99\%$, AUC-ROC ≥ 0.99 , F1-score ≥ 0.99 , consistent with the 99.60% achieved by the single ResNet-50 baseline.
- **Stage 2 (GAN vs. DM):** Ensemble accuracy $\geq 90\%$, AUC-ROC ≥ 0.93 , reflecting the greater difficulty of discriminating between two classes of synthetic content with partially overlapping spectral characteristics.
- **Audio Branch:** EER $\leq 5\%$ on the ASVspoof 2019 LA evaluation set, with AUC-ROC ≥ 0.97 , benchmarked against Fathan et al. [21] and CNN-LSTM [24].
- **XAI Output:** Qualitative demonstration that ensemble Grad-CAM and FFT heatmaps consistently distinguish the visual and spectral signatures of real, GAN-generated, and DM-generated images.
- **Platform:** A functional Gradio-based web application returning detection results with full explainability panels within a target latency of 5–10 seconds per query on Google Colab hardware.

Table 2: Image Dimensions per Forensic Category

Category	Resolution	Count	
0	DMs	244x244	330
1	DMs	256x256	170
2	GANs	200x200	121
3	GANs	244x244	138
4	GANs	256x256	241
5	Real	256x256	500

 Color Space Verification

Table 3: Distribution of Color Modes across the Corpus

Category	Color Mode	Count	
0	DMs	RGB	500
1	GANs	RGB	500
2	Real	RGB	500

Fig. 4: This figure presents an analysis of the dataset images. The top table shows the distribution of image resolutions across forensic categories—Diffusion Models (DMs), GAN-generated images, and Real images—highlighting the number of samples at different resolutions (200×200, 244×244, 256×256). The bottom table verifies the color space of the dataset, indicating that all images are in RGB format, ensuring consistency for preprocessing and model training

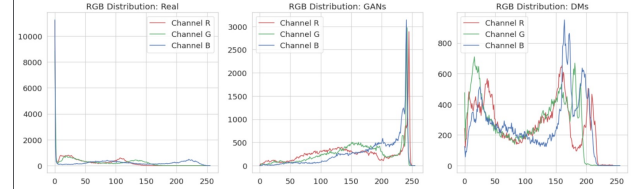


Fig. 5: RGB channel distributions for images in the dataset across three forensic categories: Real, GAN-generated, and Diffusion Model (DM) images. The plots illustrate the intensity histograms of the Red, Green, and Blue channels for each category, highlighting differences in color characteristics that can be leveraged for preprocessing, normalization, and forensic feature extraction in deepfake detection models.

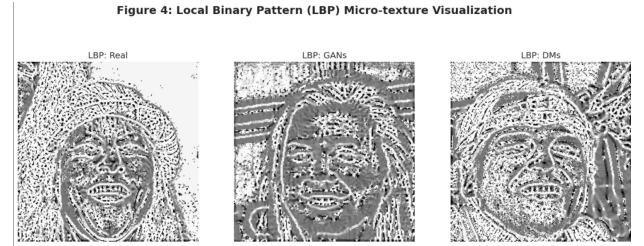


Fig. 6: Local Binary Pattern (LBP) micro-texture visualization for Real, GAN-generated, and Diffusion Model (DM) images. The plots highlight the fine-grained textural patterns present in each category, revealing structural differences that can assist in discriminating between authentic and synthetic images. LBP features emphasize edges and micro-texture irregularities, which are particularly useful for forensic feature extraction in deepfake detection models.

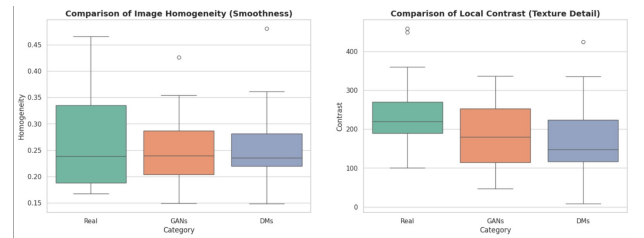


Fig. 7: Comparison of image homogeneity and local contrast for Real, GAN-generated, and Diffusion Model (DM) images. The left boxplot illustrates the smoothness (homogeneity) of images across categories, indicating that Real images tend to exhibit slightly higher homogeneity. The right boxplot shows local contrast (texture detail), highlighting the variation in micro-texture patterns, which can be leveraged for discriminative feature extraction in deepfake detection models

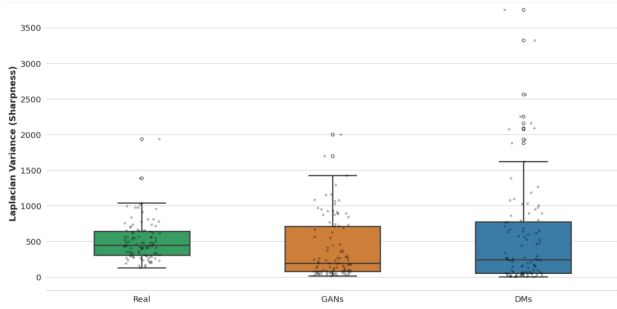


Fig. 8: Laplacian variance analysis of image sharpness for Real, GAN-generated, and Diffusion Model (DM) images. This boxplot illustrates the distribution of sharpness values across categories, highlighting that Real images generally exhibit higher sharpness compared to synthetic images. Such sharpness metrics can be exploited as discriminative features for deepfake detection.

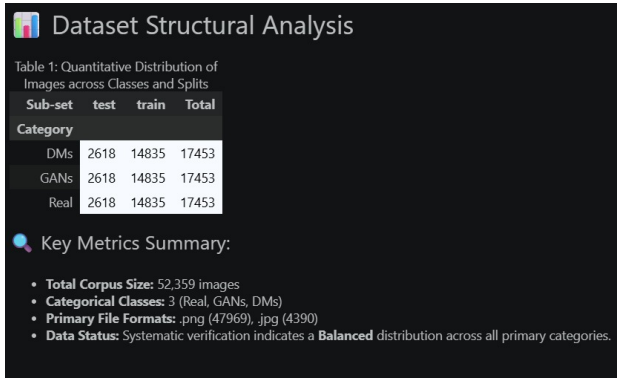


Fig. 9: Dataset structural analysis showing the quantitative distribution of images across training and test splits for each category (Real, GAN, DMs). The table highlights a balanced distribution of 52,359 images, with each category containing 17,453 images. Key metrics include the number of images per file format (.png, .jpg) and verification that all primary categories maintain balance, which is essential for unbiased model training and evaluation.

	Mean Sharpness	Std Dev	Median
Category			
DMs	555.651581	729.463118	247.089906
GANs	400.005616	424.935904	194.215025
Real	508.070034	278.175226	447.925795

Fig. 10: Statistical sharpness metrics across Real, GAN-generated, and Diffusion Model (DM) images. The table reports mean sharpness, standard deviation, and median values for each category, highlighting that Real images generally have higher median sharpness, while DMs show the highest variance. These metrics provide insight into texture and edge characteristics that can support discriminative feature extraction for deepfake detection.

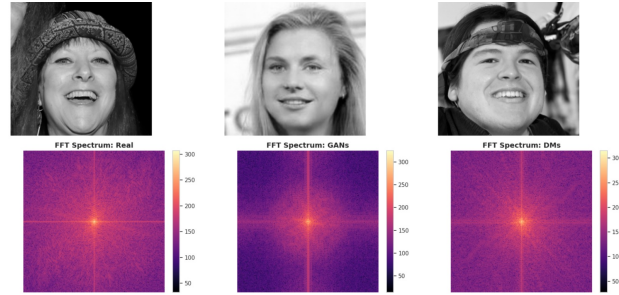


Fig. 11: FFT spectrum analysis of sample images from Real, GAN-generated, and Diffusion Model (DM) categories. The top row shows the original images, while the bottom row displays the corresponding 2-D Fast Fourier Transform (FFT) magnitude spectra. These spectral visualizations reveal periodic patterns and frequency-domain signatures characteristic of different generative processes, which can be exploited as features for deepfake detection models.

	Brightness	Sharpness	Entropy	Contrast	Homogeneity	HF_Energy
Category						
DMs	112.486	483.868	7.455	95.448	0.338	0.975
GANs	111.860	169.666	7.283	61.456	0.346	0.893
Real	119.050	508.070	7.375	132.794	0.303	1.253

Fig. 12: Mean values of key forensic features across Real, GAN-generated, and Diffusion Model (DM) images. The table compares brightness, sharpness, entropy, contrast, homogeneity, and high-frequency energy (HF_Energy) metrics, highlighting category-specific differences. These metrics provide a holistic view of discriminative properties that can aid deepfake detection and classification.

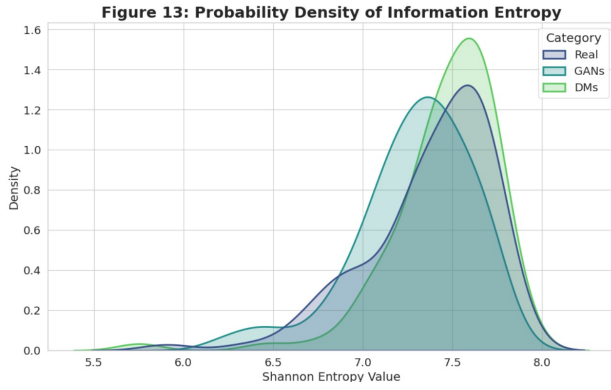


Fig. 13: Probability density of information entropy (Shannon entropy) across Real, GAN-generated, and Diffusion Model (DM) images. The plot illustrates the distribution of entropy values for each category, highlighting differences in information content and complexity. These entropy characteristics can serve as discriminative features in deepfake detection pipelines.

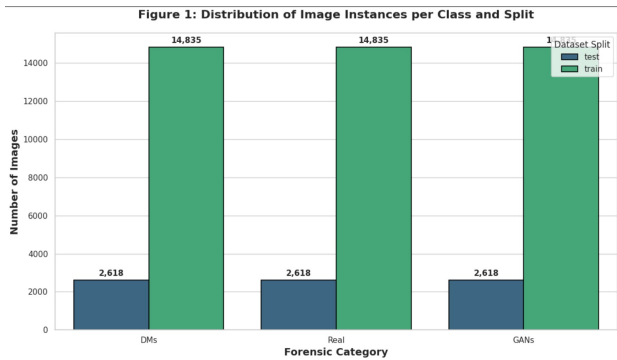


Fig. 14: Distribution of image instances across Real, GAN-generated, and Diffusion Model (DM) categories for training and testing splits. The bar chart illustrates a balanced dataset with 14,835 training images and 2,618 test images per category, ensuring uniform representation for model training and evaluation.

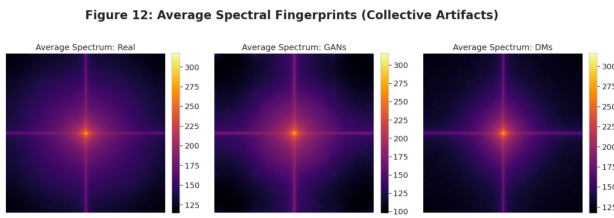


Fig. 15: Average spectral fingerprints of Real, GAN-generated, and Diffusion Model (DM) images, illustrating collective artifacts in the frequency domain. The plots show the mean FFT magnitude spectra for each category, highlighting category-specific frequency patterns that can serve as discriminative features for deepfake detection and forensic analysis.



Fig. 16: Representative sample images from the Real, GAN-generated, and Diffusion Model (DM) categories. Each row displays five sample images from a category, illustrating the diversity and visual characteristics of authentic and synthetic images. These samples provide qualitative insight into the dataset and help contextualize the forensic analyses conducted in subsequent experiments.

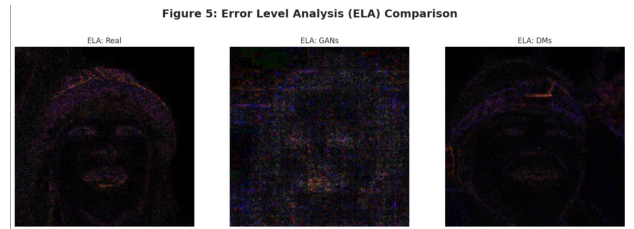


Fig. 17: Error Level Analysis (ELA) comparison across Real, GAN-generated, and Diffusion Model (DM) images. The plots highlight compression artifacts and subtle inconsistencies in the images, with synthetic images showing distinct error patterns. ELA visualizations are useful for forensic feature extraction and deepfake detection.

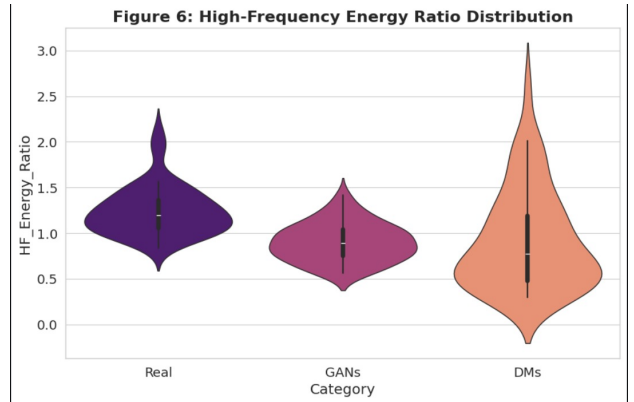


Fig. 18: Distribution of High-Frequency (HF) Energy Ratio across Real, GAN-generated, and Diffusion Model (DM) images. The violin plots highlight the spread and density of HF energy, revealing that DM images exhibit greater variation, while GANs show lower HF energy values. HF energy characteristics can be used as discriminative features for forensic analysis and deepfake detection.

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